THE BECKMAN CENTER FOR THE HISTORY OF CHEMISTRY

FRANCIS O. RICE

Transcript of an Interview
Conducted by

John A. Heitmann

at

South Bend, Indiana, and Mishawaka, Indiana

on

4 and 5 January 1984

Francis O. Rice

JH
3/15/96

CENTER FOR HISTORY OF CHEMISTRY ORAL HISTORY PROJECT

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FRANCIS O. RICE

1890	Born in Liverpool, England, 20 May

Education

1911	B.Sc., chemistry, University of Liverpool
1912	M.Sc., chemistry, University of Liverpool
1916	D.Sc., chemistry, University of Liverpool

Employment

1919	1851 Exhibition Fellow, Princeton University
	Chemistry Department, New York University
1919-1920	Instructor
1920-1924	Assistant Professor
	Chemistry Department, Johns Hopkins University
1924-1926	Associate
1926-1938	Associate Professor
1938-1959	Professor and Head of Department, Chemistry Department, Catholic University of America
1959-1962	Professor and Chairman of Department, Chemistry Department, Georgetown University
1962-1968	Visiting Research Professor and Principal Research Scientist, University of Notre Dame
1968	Principal Research Scientist, Institute for Cooperative Research, Johns Hopkins University
1968-	Director, Research Chemistry Lab, Johns Hopkins University

ABSTRACT: Francis O. Rice discusses his life and career in this interview. He begins by mentioning his early days in England, his studies at Princeton, and his teaching at New York University. The interview elucidates Rice's teaching, research, and administrative activities at the Johns Hopkins university and the Catholic University of America. Of central concern is Rice's theory of free radicals. Mrs. Katherine Rice contributes to the interview by discussing her husband's professional activities and remembering several of his closest colleagues. The interview concludes with an appraisal of the place of science in Catholic universities and an explanation of the Laidler-ADX controversy of the mid-1950s.

INTERVIEWER: John A. Heitmann holds a bachelor's degree in chemistry from Davidson College and a master's degree in history from Clemson University. From 1974 to 1979 he worked as a chemist in the metallurgical industry. He then studied at the Johns Hopkins University under Owen Hannaway and received his doctorate in the history of science in 1983.

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INTERVIEW: Francis O. Rice
Other participants: Mrs. Katherine Rice and Ms. Joan Tweedell (his personal secretary)

INTERVIEWED BY: John A. Heitmann

PLACE: South Bend, Indiana, and Mishawaka, Indiana

DATE: January 4 and 5, 1984

Heitmann: I'm with Dr. Francis O. Rice in South Bend,Indiana. I'm going to interview him today in his office and tomorrow at his home. Dr. Rice, I know that you were born in 1890 in Liverpool. Who were your parents and did you have brothers and sisters?

Rice: I come of Irish stock. My father was born in the north of Ireland; my mother was born in the south of Ireland. They met in Liverpool where I was born and educated. As far as I'm concerned I'm English. My father's people were farmers in Ireland. Due to troubled times there they emigrated, as did my father. He had some job on the railway in Liverpool. After he retired my parents moved to the north of Ireland.

Heitmann: Was he educated in a university?

Rice: He had very little education. He recognized that I had an interest in chemistry, however, and he did all that he could to encourage me to study it. He even allowed me to have a laboratory in our home when I was a boy. It's now almost a century since those days and I have no memory at all of what I did there. I doubt it was anything of any importance at all.

Heitmann: Did your mother also encourage you to do chemistry?

Rice: No, I have no recollection that she had any knowledge of it.

Heitmann: Do you have brothers and sisters?

Rice: I was the eldest in the family. I have a younger brother if he's alive. I've lost touch with him. His inclination was towards mathematics. He taught mathematics in a high school in Manchester. I also have two sisters. One is married and one is not. They live in our home, or what was our home, in Ireland.

Heitmann: Where in Ireland?

Rice: About six miles south of Armagh which is the capital of

Northern Ireland, I think. It's a historic little town. Our home was right on the border between Ulster and the Free State, although technically in Ulster. I think I have a picture someplace of the house.

Heitmann: How did you get interested in chemistry at such an early age? Were there any magazines that you read or was it just the times?

Rice: I simply don't remember. In the high school we had a course in chemistry taught by a man named Udall. I was a prize student.

Heitmann: This was in Liverpool?

Rice: In Liverpool.

Heitmann: We talked about your home laboratory a bit and how your father encouraged you.

Rice: My family are farming stock and the only thing I remember at all of any kind of gift in my family was my father's work with a machine. A neighbor bought it. It was for cereal.

Heitmann: A kind of mill?

Rice: Yes. Cereal came out at one end and waste at the other. No one could get it to function until my father worked on it. That's the only thing I remember. Apparently I did inherit something of that.

Heitmann: You did all your studies at the University of Liverpool. Is there any reason why you chose to go to Liverpool?

Rice: Well, in England there were two major universities, Oxford and Cambridge. There were also four northern universities: Leeds, Manchester, Liverpool,and Sheffield. Since I lived in Liverpool and the university was in the town, I went to Liverpool. I remember looking at the papers one morning on the way to school to see which of them made the most telling statement about a great piece of news. I decided that the Liverpool Mercury had. The statement was, "Titanic Founders with 1800 Souls"--nothing more than that.

Heitmann: Who was your chemistry professor at Liverpool?

Rice: I think E. Baly, in spectroscopy, was my major professor. A man by the name of Henry Bassett was also there, as was F. G. Donnan.

Heitmann: What type of research did you do there?

Rice: Oh dear, I don't remember.

Rice: I wrote a letter to him advising him to write a history of chemistry. I can hardly believe that Laidler would still endorse ADX .

Heitmann: Would you say that that was about the most difficult administrative problem you had in your career?

Rice: Yes. It was a very difficult period.

Heitmann: There is a question I wanted to ask you. Your academic career spanned fifty-one years. Looking back, can you think of any central achievement which you think transformed chemistry during your own lifetime?

Rice: Well.

Heitmann: Linus Pauling, do you think?

Rice: I really know too little about his work, but I came to America in 1919. I was at New York University, then Hopkins, and then Catholic University. I was a year or two at Georgetown; then I came up to Notre Dame. I'm here now steadily working along.

Heitmann: What did you think of the work of Henry Eyring? He never won a Nobel Prize.

Rice: No. Very knowledgeable man in mathematics.

Heitmann: You say the idea of speed of reactions and chemical kinetics did a great deal to transform what we know as chemistry today.

Rice: I think that's right. The speed of chemical reactions. The mechanism of chemical reactions.

Heitmann: That was all new when you first started your chemical career.

Rice: Oh, yes. Of course, when I started work, E. Baly was my professor at Liverpool and his work was mainly spectroscopic.

Heitmann: I think I'll close the taping for now. Thank you for the interview, Dr. Rice.

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